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# A Introduction

Goal of the work, maybe a simple example, some mention to preliminary information and a soft mention to what's the problem and the philosophy.

# B Minkowski space and space-like surfaces

The environment we will work within is the Minkowki 3-space, defined as $\mathbb{R}^3$ with the pseudometric

$$ds^2 = (dx^1)^2 + (dx^2)^2 - (dx^3)^2.$$

In order to distinguish this space from $\mathbb{R}^3$ we will use the notation $\mathbb{M}^{2,1}$. We will also denote the coordinates as $(x^1, x^2, x^3)$ or equivalently $(x, y, t)$. Aggiungere definizione di space-like, time-like e light-like

**Space-like immersions.**
The object we will focus on are the so-called space-like surfaces of $\mathbb{M}^{2,1}$. This term refers to the immersed surfaces in $\mathbb{M}^{2,1}$ whose induced metric is positive-definite hence Riemannian.

A nice property of this kind of surface is that they are locally the graph of a function of the form $x^3 = f(x^1, x^2)$.
This fact can be seen with a simple reasoning:
First, we assumed that the pseudometric restricted to $S$,$\langle \cdot, \cdot \rangle |_S \geq 0$, is definite positive, which means that the tangent vectors of $S$ at any point $p$ belong to the set of space-like vectors:

$$T_p \subset \mathscr{C} := \left\{ v \in \mathbb{M}^{2,1} : \langle v, v \rangle \geq 0 \right\} = \{ t^2 \leq x^2 + y^2 \}.$$

This is just the outside-region of a double cone in $\mathbb{M}^{2,1}$, whose symmetry axis is $\left\{ x^1 = 0, x^2 = 0 \right\}$. A plane is entirely made of space-like vectors iff it's contained between the two branches of the cone.

Second, having assumed that $S$ is an immersed surface, hence an immersed submanifold, it is locally the solution of an equation of the form $F(x, y, z) = 0$. It is well known that from this local expression we can express the normal vector to the surface at a point $p$: $\vec{n}_p = \vec{\nabla} F(p)$.
Furthermore, we observe that, being $\vec{n}_p$ orthogonal to $T_pS$, it cannot be horizontal, $(\vec{n}_p)^3 = \partial_z F(p) \neq 0$, which means that we can apply the implicit function theorem and express the surface as the graph of $z = f(x, y)$.
Furthermore, where the local representation holds, we can parametrize the surface as $(x, y, f(x, y))$, from which two generators of the tangent space are $V_x = (1, 0, \partial_x f(x, y))$ and $V_y = (1, 0, \partial_y f(x, y))$, and the first fundamental form is represented by the matrix

$$g = \begin{pmatrix} \langle V_x, V_x \rangle & \langle V_y, V_x \rangle \\ \langle V_x, V_y \rangle & \langle V_y, V_y \rangle \end{pmatrix} = \begin{pmatrix} 1 - f_x^2 & -f_x f_y \\ -f_x f_y & 1 - f_y^2 \end{pmatrix}.$$

The condition on this being a metric translates in having a strictly positive determinant (as the parametrization is obviously regular),

$$0 < \det g = 1 - (f_x^2 + f_y^2) = 1 - |\vec{\nabla} f|^2,$$

which means $|\vec{\nabla} f|^2 < 1$. It's not difficult to convince ourselves, as the local representation as a graph is always possible, that the conditions of being able to explicit the time variable as dependent and that $|\vec{\nabla} f|^2 < 1$ are a characterization of space-like immersed surfaces.mettere dopo

**Entire surfaces.**
Given a surface $S$ immersed in $\mathbb{M}^{2,1}$, we can define a projection $\Pi : S \to \mathbb{R}^2$ by forgetting the time coordinate, $\Pi(x^1, x^2, x^3) := (x^1, x^2)$ for any $(x^1, x^2, x^3) \in S$. We call the surface $S$ is entire when the map $\Pi$ is surjective. If the surface is globally a graph of a function $\varphi$, this condition is exactly stating that the function $\varphi$ is an entire map from $\mathbb{R}^2$ to $\mathbb{R}$.

The situation for space-like surfaces is actually a lot nicer than what we presented up to now, as the representation as a graph is not only local but actually global, as follows from the proposition below. manca mostrare che davvero la rappresentazione è globale ovvero la proposizione seguente

**Proposition 1.** *Let $\Omega$ be a domain in $\mathbb{C}$ (that is, an open connected set) and let $S : \Omega \to \mathbb{M}^{2,1}$ be a space-like immersion. Suppose that $S$ is complete with respect to the induced Riemannian metric. Then $S$ is an entire graph and $\Omega$ is simply connected.*

**Immersion data for $S$**
As we already mentioned, the immersion $\iota : S \hookrightarrow \mathbb{M}^{2,1}$ induces both a scalar product on $TS$, the first fundamental form or pull-back metric, which we indicate as $\mathbf{I} = \iota^*(g_{\mathbb{M}^{2,1}})$ and in coordinates $\mathbf{I} = g_{ij} dx^i dx^j$. Here we are identifying the involved and their composition with the immersion, $x^i = x^i \circ \iota$, $g_{ij} = g_{ij} \circ \iota$. Analogously we indicate the second fundamental form as $\mathbf{II} = h_{ij} dx^i dx_j$. Later we will look at the second fundamental form more deeply as it's involved in the definition of the Hopf differential.

**The Gauss map in $\mathbb{M}^{2,1}$**

The surface $S$ is an immersed submanifold of codimension 1. As we have a non-degenerate form on the ambient space, $g$, once fixed a $p \in S$, we can decompose the tangent spaces as follows:

$$\mathbb{M}^{2,1} = T_p\mathbb{M}^{2,1} = (T_pS)^\perp \oplus T_pS =: P^\perp \oplus P.$$

Having codimension 1, the dimension of $(T_pS)^\perp$ is exaclty one, and in fact it's generated only by the normal vector to $S$, $\vec{n}$.

We can easily show that $n$ must be a time-like vector, due to the fact that $T_pS$ is made of space-like vectors.

Let's fix $e_1, e_2 \in T_pM$ two independent. As $n$ belongs to $(T_pM)^\perp$ we have $\langle e_1, n\rangle = \langle e_2, n\rangle = 0$. This implies that when we look at the representative matrix of $g_p$, we have

$$[g_p] = \begin{pmatrix} \langle n, n\rangle & 0 & 0 \\ 0 & \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ 0 & \langle e_2, e_1\rangle & \langle e_2, e_2\rangle. \end{pmatrix} =: \begin{pmatrix} \langle n, n\rangle & 0 \\ 0 & [(g|_S)_p] \end{pmatrix}.$$

Where the block $[(g|_S)_p]$ represents the form $g$ on the subspace $T_pM$. We conclude that $\langle n, n\rangle$ is negative just by observing that $\det[g_p] < 0$ as $g$ is a pseudometric, while $\det[(g|_S)_p] > 0$ since $g$ becomes a metric when restricted to $S$.

We can now repeat the classical construction of the Gauss map in this pseusoriemannian context.

The main idea is to take a unit vector in $(T_pM)^\perp$, as we have just seen we can't impose the squared norm to be 1, which means we need to normalize it to $-1$. Given $w \in (T_pM)^\perp$ we scale by a factor $k$ such that $\langle kw, kw\rangle = -1$. This equation has two different solutions, that send $w$ to one of the two branches of $\hat{\mathbb{H}}^2 = \{v \in \mathbb{M}^{2,1} |\, \langle v, v\rangle = -1\}$, so we need to make a choice. We consistently choose it to be in $\mathbb{H}^2 = \{v \in \mathbb{M}^{2,1} |\, \langle v, v\rangle = -1, v^3 > 0\}$.

This way we have constructed a map

$$u : S \to \mathbb{H}^2,$$

the Gauss map of the surface $S$. Of course the only difference from the classical theory of surfaces is the codomain of the map $u$, where it is the sphere $\mathbb{S}^2$, and like in the classical case, the of which normal vector to choose specifies an orientation for the surface.

$\mathbb{H}^2$ is a model for the Hyperbolic space.

**The curvatures of a space-like map**

- $\mathbf{M}^{2,1}$
- Space-like immersion, induced Riemannian metric
- First and second fundamental forms
- Mean curvature and Gaussian curvature expr
- Space-like surfaces are graphs of functions
- The Gauß map has $\mathbb{H}^2$ as a target

# C Harmonic maps

# D Classification of CMC

## D.1 The idea of the proof

## D.2 The analytical work